

$$\textcircled{1} \Omega = \{\omega_2; \omega_4; \omega_6; \hat{\omega}_1; \hat{\omega}_3; \hat{\omega}_5\}$$

$$a) \omega_i := \text{Die shows } i; \hat{\omega}_j = \{\omega_{j1}; \omega_{j2}; \omega_{j3}; \omega_{j4}; \omega_{j5}; \omega_{j6}\}$$

$\omega_{ji} := \text{First, } j \text{ was rolled, and then } i \text{ was rolled}$

$$\Omega = \{\omega_2; \omega_4; \omega_6; \{\omega_{11}; \omega_{12}; \omega_{13}; \omega_{14}; \omega_{15}; \omega_{16}\}; \\ ; \{\omega_{31}; \omega_{32}; \omega_{33}; \omega_{34}; \omega_{35}; \omega_{36}\}; \\ ; \{\omega_{51}; \omega_{52}; \omega_{53}; \omega_{54}; \omega_{55}; \omega_{56}\}\}$$

$$|\Omega| = 3 + 3 \cdot 6 = 21$$

$$b) P(\text{result} : 6) - ?$$

$$\text{result} : 6 := \{\omega_6; \omega_{15}; \omega_{33}; \omega_{51}\}; \omega_6 \cap \omega_{15} \cap \omega_{33} \cap \omega_{51} = \emptyset$$

$$P(\text{result} : 6) = P(\omega_6) + P(\omega_{15}) + P(\omega_{33}) + P(\omega_{51}) = \\ = \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{4}$$

$$c)$$

ξ	2	3	4	5	6	7	8	9	10	11
P	$\frac{7}{36}$	$\frac{1}{36}$	$\frac{2}{9}$	$\frac{1}{18}$	$\frac{1}{4}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{36}$	$\frac{1}{36}$

ξ	1	2	3	4	5	6	7	8	9	10	12	15	18	20	25	30
P	$\frac{1}{36}$	$\frac{7}{36}$	$\frac{1}{18}$	$\frac{7}{36}$	$\frac{1}{18}$	$\frac{2}{9}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

$$E(x) = \sum_{i=1}^n P_i x_i; D(x) = E(x^2) - (E(x))^2$$

$$E_{\xi} = \frac{21}{4}; E_{\xi}(x^2) = 33.25; D_{\xi} = 33.25 - (5.25)^2 = 5.6875$$

$$E_Y = 7.25 \quad E_Z(x^2) \approx 97.81 \quad D_Z = 97.81 - (7.25)^2 \approx 45.24$$

d) $(\bar{Z}, \bar{Y}) = ? ; \text{cov}(\bar{Z}, \bar{Y})$

$\bar{Y} \backslash \bar{Z}$	1	2	3	4	5	6	7	8	9	10	11	12	15	18	20	25	30
2	$\frac{1}{36}$	$\frac{1}{6}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	$\frac{1}{18}$	$\frac{1}{6}$	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	$\frac{1}{36}$	0	$\frac{1}{36}$	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	$\frac{1}{18}$	$\frac{1}{6}$	$\frac{1}{36}$	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	$\frac{1}{36}$	0	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	$\frac{1}{18}$	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	$\frac{1}{36}$	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	$\frac{1}{36}$

$$\text{cov}(\bar{Z}, \bar{Y}) = E[\bar{Z}\bar{Y}] - E_{\bar{Z}} E_{\bar{Y}}$$

$$E_{\bar{Z}\bar{Y}} = \sum_i \sum_j x_i y_j P_{ij} = 174.4375 \quad 52.5$$

$$\text{cov}(\bar{Z}, \bar{Y}) = 174.4375 - 7.25 \cdot 52.5 = 14.4375$$

$$\text{cov}(\bar{Z}, \bar{Y}) > 0 \Rightarrow \bar{Z} \text{ \& } \bar{Y} \text{ dependent, } P(\bar{Z}, \bar{Y}) \neq P_{\bar{Z}} \cdot P_{\bar{Y}}$$

② $N = N_1 + N_2 + N_3 + N_4$

Two students were selected: A - first; B - second
 $A > B$:= A has been studying longer than B

$A \in N_i$:= student A is in i -th

$P(A \in N_3 / A > B) = ?$

First solution:

~~$P(A > B / A \in N_3) =$~~

$$P(A \in N_3 / A > B) = \frac{P((A \in N_3) \cap (A > B))}{P(A > B)}$$

$$P((A \in N_3) \cap (A > B)) = \frac{n_3(n_2 + n_1)}{N(N-1)}$$

$$P(A > B) = \frac{\sum_{1 \leq i < j \leq 4} n_j n_i}{N(N-1)} = \frac{n_4(n_3 + n_2 + n_1) + n_3(n_2 + n_1) + n_2 n_1}{N(N-1)}$$

$$P(A \in N_3 / A > B) = \frac{n_3(n_2 + n_1)}{n_3(n_2 + n_1) + n_4(n_3 + n_2 + n_1) + n_2 n_1}$$

Second solution:

$$P(A \in N_3 / A > B) = \frac{P(A > B / A \in N_3) P(A \in N_3)}{\sum_k P(A > B / A \in N_k) P(A \in N_k)}$$

$$P(A > B / A \in N_k) = \frac{\sum_{j=1}^{k-1} n_k n_j}{n_k(N-1)} = \frac{\sum_{j=1}^{k-1} n_j}{N-1}$$

$k \neq 1$

$$P(A \in \mathcal{N}_k) = \frac{n_k(n-1)}{n(n-1)} = \frac{n_k}{n}$$

$$P(A \in \mathcal{N}_3 | A > B) = \frac{(n_2 + n_1)}{(n-1)} \cdot \frac{n_3}{n} = \frac{\frac{(n_3 + n_2 + n_1)}{(n-1)} \cdot \frac{n_4}{n} + \frac{(n_2 + n_1)}{(n-1)} \cdot \frac{n_3}{n} + \frac{n_1}{(n-1)} \cdot \frac{n_2}{n}}{n_4(n_3 + n_2 + n_1) + n_3(n_2 + n_1) + n_1 n_2}$$

new = 9
old = 15 - 9 = 6

③ $\rightarrow P(A/B) = ?$
 $B :=$ taken 3 balls before
 $AB :=$ taken 3 balls after new

$$P(A/B) = P(A/\text{new}=9) \cdot P(\text{new}=9) + P(A/\text{new}=8) \cdot P(\text{new}=8) + P(A/\text{new}=7) \cdot P(\text{new}=7) + P(A/\text{new}=6) \cdot P(\text{new}=6)$$

new = k := there are k new balls after first taking three

$$P(A/B) = \frac{\binom{9}{3}}{\binom{15}{3}} \cdot \frac{\binom{6}{3}}{\binom{15}{3}} + \frac{\binom{8}{3}}{\binom{15}{3}} \cdot \frac{\binom{6}{2} \cdot \binom{9}{1}}{\binom{15}{3}} + \frac{\binom{7}{3} \cdot \binom{6}{1} \cdot \binom{9}{2}}{\binom{15}{3} \cdot \binom{15}{3}} + \frac{\binom{6}{3} \cdot \binom{9}{3}}{\binom{15}{3} \cdot \binom{15}{3}} = \left(\frac{1}{5 \cdot 7 \cdot 13} \right)^2 \cdot [84 \cdot 20 + 56 \cdot 15 \cdot 9 + 35 \cdot 6 \cdot 36 + 20 \cdot 84] = \frac{18480}{207025} \approx 0.09$$

4) $f_3(x) = \text{const} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$

a) const - ?

$$\begin{aligned} \int_{-\infty}^{+\infty} f_3(x) dx &= 1 = \text{const} \int_{-\infty}^{+\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \text{const} \int_{-\infty}^{+\infty} \exp\left(-\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)^2\right) dx = \\ &= \text{const} \sqrt{2}\sigma \int_{-\infty}^{+\infty} \exp\left(-\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)^2\right) d\left(\frac{x-\mu}{\sqrt{2}\sigma}\right) = \text{const} \sqrt{2}\sigma \cdot \sqrt{\pi} = 1 \Rightarrow \\ &\Rightarrow \text{const} = \frac{1}{\sqrt{2\pi}\sigma^2} \end{aligned}$$

c) $\chi_3(x) = ?$

$$\begin{aligned} \chi_3(x) &= \int_{-\infty}^{+\infty} f_3(t) e^{itx} dx \Rightarrow \frac{1}{\sqrt{2\pi}\sigma^2} \int_{-\infty}^{+\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{itx} dx = \\ &= \frac{1}{\sqrt{2\pi}\sigma^2} \int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2\sigma^2} \left[x^2 - 2x\left(\frac{2\mu - 2\sigma^2 it}{2}\right) + \mu^2 \right]\right) dx = \frac{1}{\sqrt{2\pi}\sigma^2} \cdot \\ &\cdot \int_{-\infty}^{+\infty} \exp\left(-\frac{1}{2\sigma^2} \left[x - \left(\frac{2\mu - 2\sigma^2 it}{2}\right)^2 \right]\right) \cdot \exp\left(-\frac{1}{2\sigma^2} \left[\frac{2\sigma^4 t^2}{2} - 2\sigma^2 \mu it \right]\right) \cdot \\ &\cdot d\left(\frac{x - \left(\frac{2\mu - 2\sigma^2 it}{2}\right)^2}{\sqrt{2\sigma^2}}\right) \cdot \sqrt{2\sigma^2} = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-x^2} dx \cdot \exp\left(\mu it - \frac{\sigma^2 t^2}{2}\right) = \\ &= \exp\left(\mu it - \frac{\sigma^2 t^2}{2}\right) \end{aligned}$$

b) $i^n E_3(x^n) = \chi_3^{(n)}(t) \Big|_{t=0}$

$$E_3 = \frac{1}{i} \chi_3'(t) \Big|_{t=0} = \frac{1}{i} \cdot e^{(\mu it - \frac{\sigma^2 t^2}{2})} \cdot (\mu i - \sigma^2 t) \Big|_{t=0} = \mu$$

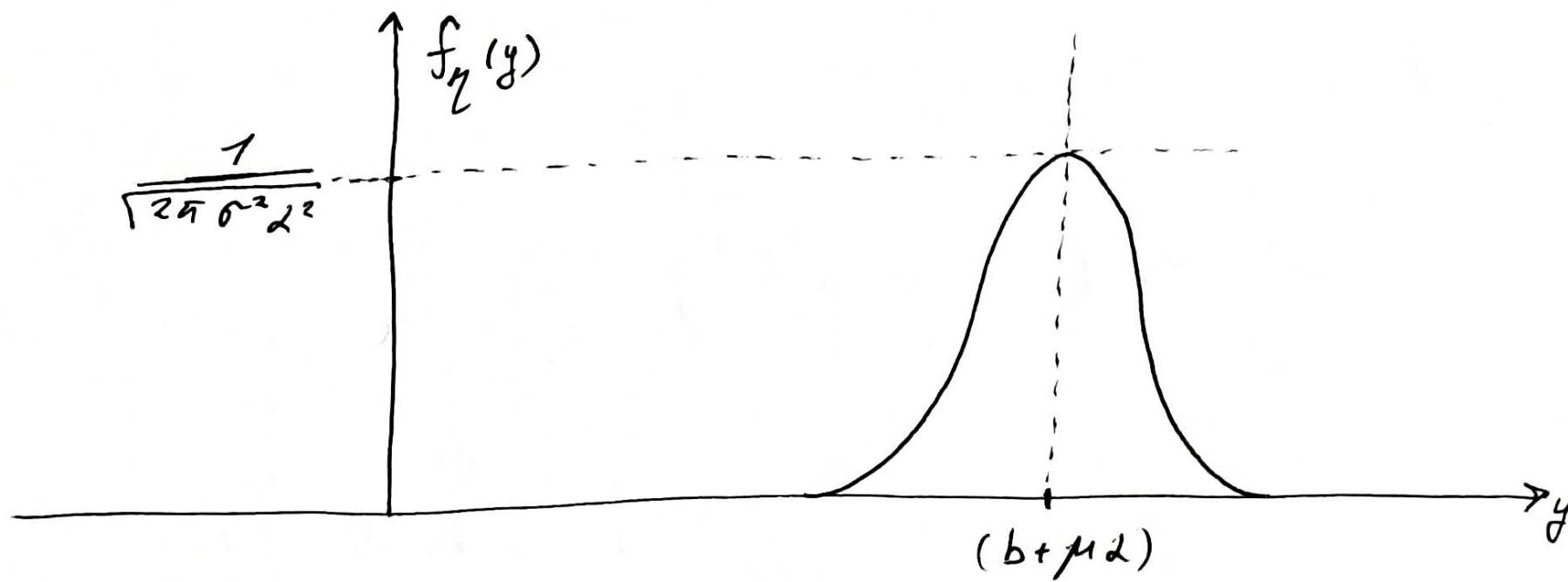
$$\begin{aligned} D_3 = E_3(x^2) - E_3^2 &= \frac{1}{i^2} e^{(\mu it - \frac{\sigma^2 t^2}{2})} (\mu i - \sigma^2 t)^2 + \frac{1}{i^2} e^{(\mu it - \frac{\sigma^2 t^2}{2})} (-\sigma^2) \Big|_{t=0} - \mu^2 = \\ &= \frac{(\mu i)^2}{i^2} + \frac{i^2 \sigma^2}{i^2} - \mu^2 = \sigma^2 \end{aligned}$$

⑤ 1) $X \sim \mathcal{N}(\mu, \sigma^2) \rightarrow f_{j_i}(x) = \frac{1}{\sqrt{2\pi}\sigma_j} \cdot \exp\left(-\frac{(x-\mu_j)^2}{2\sigma_j^2}\right)$
 $\chi_{j_i}(t) = \exp\left(i\mu_j t - \frac{\sigma_j^2 t^2}{2}\right)$

2) $S = \sum_{j=1}^N \tilde{z}_j$
 $P_{\tilde{z}_j} = \prod_{j=1}^N P_{j_i} \Rightarrow f_S(\vec{x}) = \prod_{j=1}^N f_{j_i}(x_j)$

3) $\chi_S(t) = E_S(e^{iSt}) = E_S(e^{it \sum_{j=1}^N \tilde{z}_j}) = \prod_{j=1}^N \left(E_{\tilde{z}_j}(e^{it \tilde{z}_j}) \right) =$
 $= \prod_{j=1}^N \exp\left(i\mu_j t - \frac{\sigma_j^2 t^2}{2}\right) = \exp\left(it \left(\sum_{j=1}^N \mu_j\right) - \frac{t^2}{2} \left(\sum_{j=1}^N \sigma_j^2\right)\right) =$
 $= \exp\left(it \tilde{\mu} - \frac{t^2}{2} \tilde{\sigma}^2\right) \Rightarrow S \sim \mathcal{N}(\tilde{\mu}, \tilde{\sigma}^2)$
 $\tilde{\mu} = \sum_{j=1}^N \mu_j ; \tilde{\sigma}^2 = \sum_{j=1}^N \sigma_j^2$

⑥ $z \sim \mathcal{N}(\mu, \sigma^2) ; y = az + b ; a < 0 ; f_z(y) = ?$
 $\xrightarrow{g} y : g(x) = ax + b = y ; x = g^{-1}(y) = \frac{y-b}{a}$
2) $f_z(x) dx = f_y(y) dy \Rightarrow f_y(y) = f_z(x) \frac{dx}{dy} = f_z(g^{-1}(y)) \cdot (g^{-1}(y))' \Rightarrow$
 $\Rightarrow f_y(y) = \frac{1}{\sqrt{2\pi}\sigma^2} \cdot \exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right) \cdot \left|\frac{1}{a}\right| \rightarrow$
 $\rightarrow f_y(y) = \frac{1}{\sqrt{2\pi}\sigma^2 |a|^2} \cdot \exp\left(-\frac{(y - (b + \mu a))^2}{2(\sigma a)^2}\right)$



7) $f \sim f_j(x) = x^n, x \in \left[-\frac{3}{2}; -\frac{1}{2}\right]$
 $z = \sin\left(\frac{\pi x}{2}\right); n \rightarrow \text{even}$

$f_z(y) = ?$

$F_z(y) = ?$

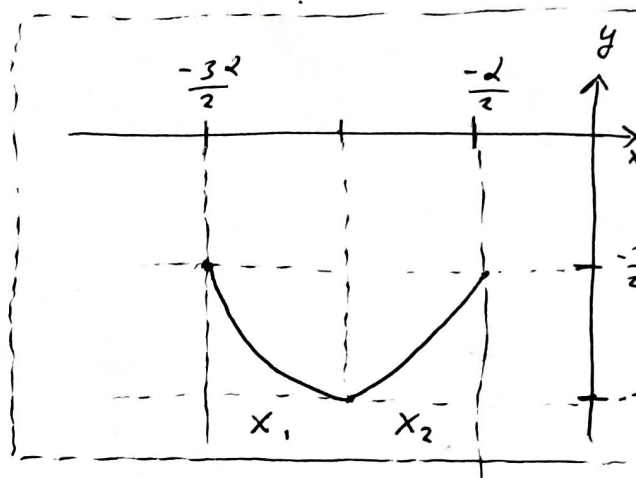
$m = ?$

1) $y = \sin\left(\frac{\pi x}{2}\right)$

$f_z(y) = \sum_{k=1}^N f_j(x_k(y)) \cdot \left| \frac{dx_k(y)}{dy} \right|$

2) $x_1(y) = \frac{2}{\pi} \cdot (-1) \cdot (\arcsin(y) + \pi)$

$x_2(y) = \frac{2}{\pi} \cdot \arcsin(y)$



$\arcsin(x) \leftrightarrow \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$

$\left| \frac{dx_1(y)}{dy} \right| = \left| \frac{dx_2(y)}{dy} \right| = \left| \frac{2}{\pi} \right| \cdot \frac{1}{\sqrt{1-y^2}}, y \in \left[-1; -\frac{1}{2}\right]$

$f_z(y) = \left| \frac{2}{\pi} \right|^{n+1} \cdot \left[(\arcsin(y) + \pi)^n + \arcsin^n(y) \right] \cdot \frac{1}{\sqrt{1-y^2}}$

3) $F_z(y) = \int_{-\infty}^y f_z(t) dt; \forall y \in \left(-1; -\frac{1}{2}\right] \Rightarrow F_z(y) = \int_{-1}^y f_z(t) dt \rightarrow$

$F_z(y) = \int_{-1}^y \left| \frac{2}{\pi} \right|^{n+1} \cdot \left[(\arcsin(t) + \pi)^n + \arcsin^n(t) \right] \cdot d(\arcsin(t)) =$
 $= \left| \frac{2}{\pi} \right|^{n+1} \cdot \frac{1}{n+1} \cdot \left[(\arcsin(y) + \pi)^{n+1} + \arcsin^{n+1}(y) - \left(\left(\frac{\pi}{2} \right)^{n+1} + \left(-\frac{\pi}{2} \right)^{n+1} \right) \right] =$

$F_z(y) = \begin{cases} 0, & y \leq -1 \\ \left| \frac{2}{\pi} \right|^{n+1} \cdot \frac{1}{n+1} \cdot \left[(\arcsin(y) + \pi)^{n+1} + \arcsin^{n+1}(y) \right], & -1 < y \leq -\frac{1}{2} \\ 1, & -\frac{1}{2} < y \end{cases}$

4) m - ?

$$F_2\left(-\frac{1}{2}\right) = 1 \Rightarrow \left(\frac{2}{\pi}\right)^{n+1} \cdot \frac{|2|^{n+1}}{n+1} \cdot \left(\left(\frac{3\pi}{4}\right)^{n+1} + \left(-\frac{\pi}{4}\right)^{n+1} \right) = 1 \rightarrow$$

$$\rightarrow |2|^{n+1} = \frac{(n+1) \cdot 2^{n+1}}{(3^{n+1} - 1)} \Rightarrow |2| = 2 \cdot \left(\frac{(n+1)}{3^{n+1} - 1} \right)^{\frac{1}{n+1}} \Rightarrow$$

$$\Rightarrow m \Rightarrow F_2(y_m) = \left(\frac{4}{\pi}\right)^{n+1} \cdot \frac{1}{(3^{n+1} - 1)} \cdot \left((\arcsin(y_m) + \pi)^{n+1} + \arcsin(y_m)^{n+1} \right) = \frac{1}{2}$$

$m = y_m$



$$F(x) = \frac{1}{2}$$



$$F_n(x) = \left(\frac{4}{\pi}\right)^{(n+1)} \cdot \frac{1}{(3^{(n+1)} - 1)} \cdot \left((\arcsin(x) + \pi)^{(n+1)} + \arcsin(x)^{(n+1)}\right)$$



$$n = 2$$

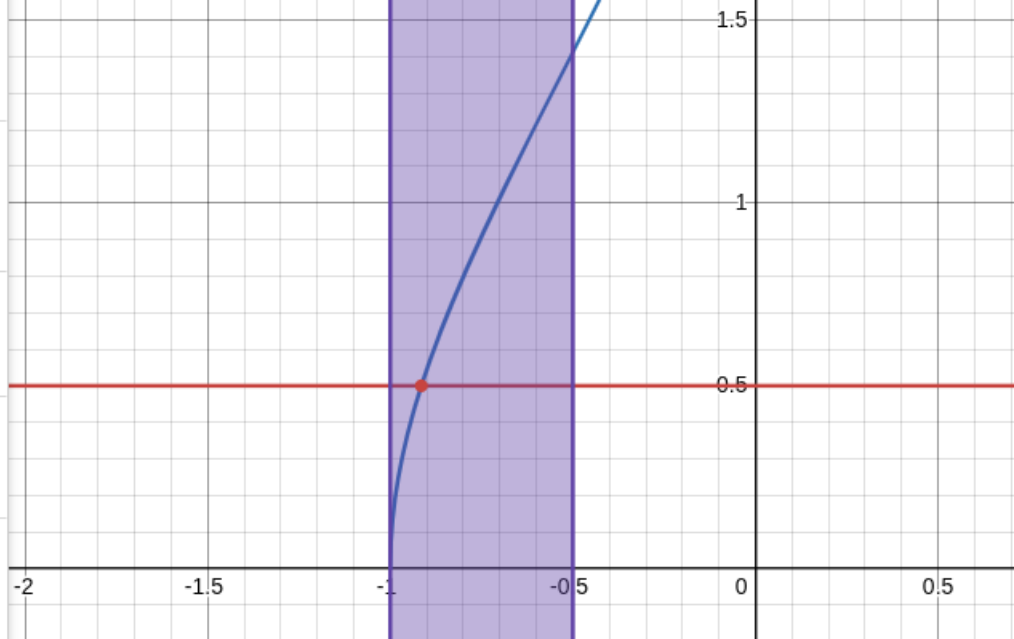


$$-1 \leq x \leq -\frac{1}{2}$$



$$(-0.91483, 0.5)$$

☐ Имя



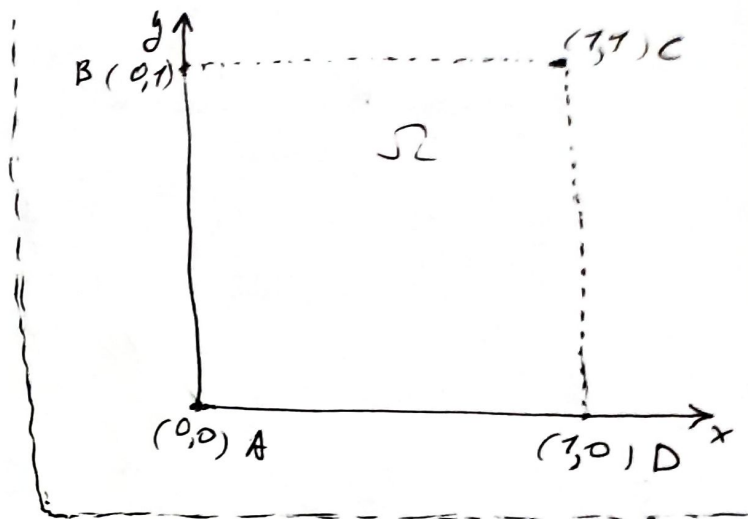
8) $f_{Z,Y}(x,y) = \text{const} \cdot (x+y)$

a) const-?

$$\int_0^1 dx \int_0^1 f_{Z,Y}(x,y) dy = 1 \Rightarrow$$

$$\text{const} \int_0^1 \int_0^1 (x+y) dx dy =$$

$$= \int_0^1 \left(x + \frac{1}{2}\right) dx \cdot \text{const} = 1 \cdot \text{const} = 1 \Rightarrow \underline{\text{const} = 1} \rightarrow f_{Z,Y}(x,y) = (x+y)$$



b) $f_Z(x) = \int_0^1 (x+y) dy = x + \frac{1}{2}$; $f_Y(y) = \int_0^1 (x+y) dx = y + \frac{1}{2}$

$$f_Z(x) \cdot f_Y(y) = \left(x + \frac{1}{2}\right) \left(y + \frac{1}{2}\right) = \left(xy + \frac{x+y}{2}\right) \neq x+y \neq f_{Z,Y}(x,y) \Rightarrow$$

$\Rightarrow Z$ and Y dependent

c) $E_Z(x) = \int_0^1 x \left(x + \frac{1}{2}\right) dx = \frac{7}{12}$; $E_Y(y) = \int_0^1 y \left(y + \frac{1}{2}\right) dy = \frac{7}{12}$

$$E_Z(x^2) = \int_0^1 x^2 \left(x + \frac{1}{2}\right) dx = \frac{5}{12} ; E_Y(y^2) = \int_0^1 y^2 \left(y + \frac{1}{2}\right) dy = \frac{5}{12}$$

$$D_Z = E_Z(x^2) - E_Z^2(x) = \frac{11}{12 \cdot 12} ; D_Y = E_Y(y^2) - E_Y^2(y) = \frac{11}{144}$$

d) r_{ZY} -? ; $r_{ZY} = \frac{\text{COV}(Z,Y)}{\sqrt{D_Z D_Y}}$; $\text{COV}(Z,Y) = E_{ZY}(xy) - E_Z(x) E_Y(y)$

$$E_{ZY}(xy) = \int_0^1 \int_0^1 xy(x+y) dx dy = \frac{1}{3}$$

$$r_{ZY} = \frac{\frac{1}{3} - \frac{7}{12} \cdot \frac{7}{12}}{\sqrt{\frac{11}{144} \cdot \frac{11}{144}}} = -\frac{1}{11} ; r_{ZY} \neq 0 \Rightarrow \text{correlated}$$

$$e) E(z|y=y) = \int_0^1 x f_z(x|y=y) dx \quad ? ; \quad \tau(y) = E(x|y=y)$$

$$f_z(x|y=y) = \frac{f_{z,y}(x,y)}{f_y(y)} = \frac{(x+y)}{y+\frac{1}{2}} \cdot \frac{2(x+y)}{2y+1}$$

$$\tau(y) = E(x|y=y) = \int_0^1 x \cdot \frac{2(x+y)}{2y+1} dx = \frac{y+\frac{2}{3}}{2y+1} \cdot \frac{3y+2}{6y+3}$$

$$\tau(y) = \frac{3y+2}{6y+3} ; \quad \tau(x) = \frac{3x+2}{6x+3}$$

$$f) Z_c(x) = b + ax \quad ; \quad a, b = \text{argmin } E[(y - (ax + b))^2]$$

$$\begin{cases} a = \frac{\text{Cov}(z, y)}{D_z} \\ b = E_z(y) - a E_z(x) \end{cases} \Rightarrow \begin{aligned} a &= \frac{-\frac{1}{144}}{\frac{11}{144}} = -\frac{1}{11} \\ b &= \frac{7}{12} + \frac{1}{11} \cdot \frac{7}{12} = \frac{7}{11} \end{aligned}$$

$$Z_c(x) = \frac{7}{11} - \frac{1}{11} x.$$



$A(0,0), B(0,1), C(1,1), D(1,0)$

☒ Имя:

=



$B = 1$

-10 10



$C = 1$

-10 10



$D = 1$

-10 10



$A = 1$

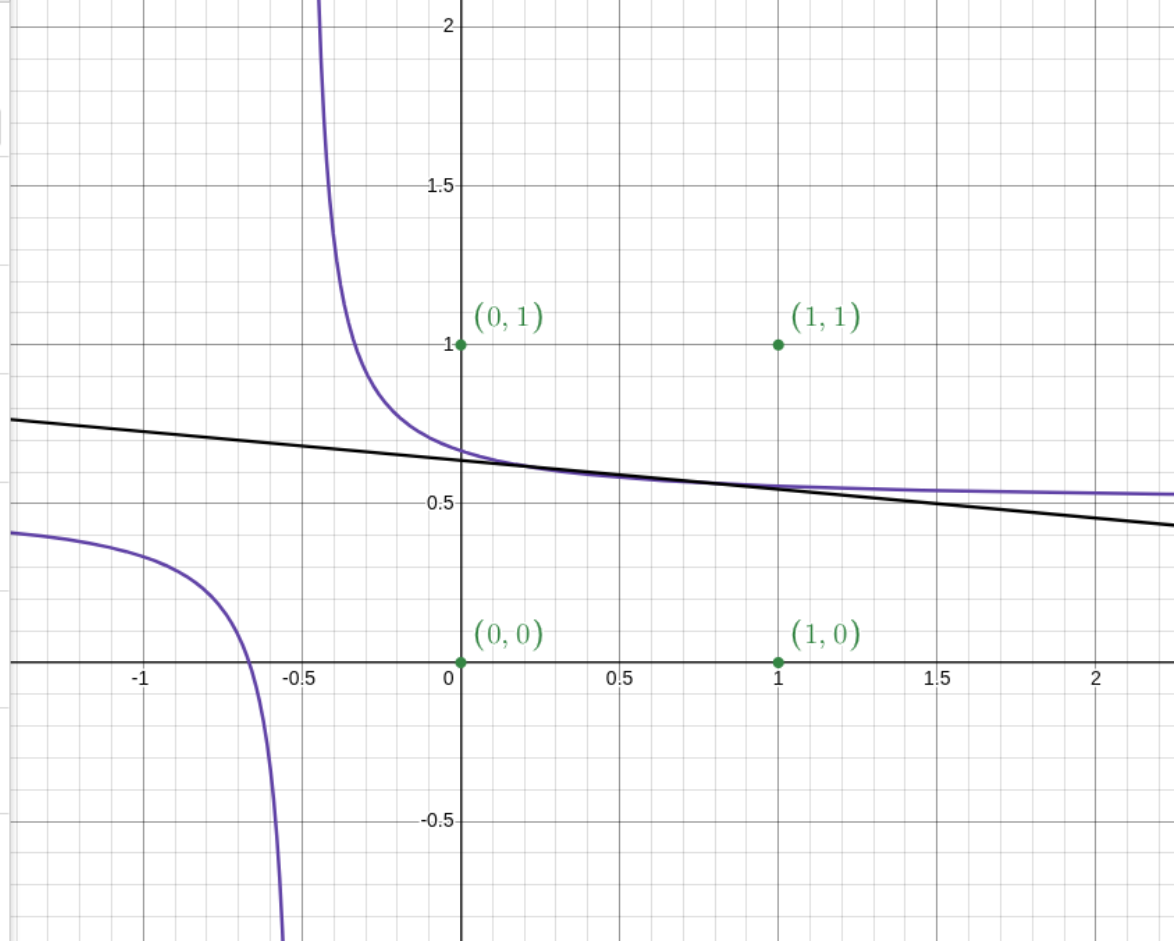
-10 10



$$r(x) = \frac{(3x+2)}{(6x+3)}$$



$$r_l(x) = \frac{7}{11} - \frac{x}{11}$$



$$\textcircled{9} \quad P(X_n = e^{-n \ln 2}) = \frac{1}{2} = P(X_n = e^{n \ln 2})$$

$$P(\lim_n \bar{X}_n = \mu) = 1$$

$$\bar{X}_n \xrightarrow[n]{\text{н.ч.}} \mu$$

$$\Delta \quad S_n = X_1 + X_2 + \dots + X_n \Rightarrow \frac{S_n}{n} \xrightarrow{\text{н.ч.}} \mu \Rightarrow \frac{E[S_n]}{n} \rightarrow E[\mu]$$

$$E[S_n] = \frac{1}{2}(2^{-1} + 1 \cdot 2^1) + \frac{1}{2}(2^{-2} + 1 \cdot 2^2) + \dots + \frac{1}{2}(2^{-n} + 1 \cdot 2^n) \Rightarrow$$

$$\frac{E[S_n]}{n} = \frac{1}{2} \sum_{k=1}^n \frac{2^{-k} + 1 \cdot 2^k}{n} \leq \frac{1}{2} \sum_{k=1}^n \frac{2^{-k}}{k} + \frac{1 \cdot 2^k}{k} \xrightarrow{n \rightarrow \infty} \infty$$

$$\frac{E[S_n]}{n} = \frac{1}{2} \sum_{k=1}^n \frac{2^{-k}}{n} + \frac{1 \cdot 2^k}{n} \xrightarrow{n \rightarrow \infty} \infty$$

$$\frac{E[S_n]}{n} \xrightarrow[n \rightarrow \infty]{\text{н.ч.}} E[\mu] = \mu < \infty$$

?! 

Невыполнение 3.5.4.

```
import numpy as np
```

```
SEED_CONST = np.uint32(17)
A_CONST = np.uint32(1664525)
C_CONST = np.uint32(10139904223)
M_CONST = np.uint64(2**32)
```

```
class UniformGenerator:
```

```
    def __init__(self, seed=None, a=None, c=None, m=None):
        self._seed = np.uint32(seed) if (seed is not None) else SEED_CONST
        self._a = np.uint32(a) if (a is not None) else A_CONST
        self._c = np.uint32(c) if (c is not None) else C_CONST
        self._m = np.uint64(m) if (m is not None) else M_CONST
        self._cur_sample = self._seed

    def _get_next_sample(self):
        self._cur_sample = (self._a * self._cur_sample + self._c) % self._m

        return self._cur_sample / self._m

    def generate_samples(self, n):
        samples = np.empty(n, dtype=np.float64)

        for i in range(n):
            samples[i] = self._get_next_sample()

        return samples

    def reset(self, seed=None):
        self._seed = np.uint32(seed) if (seed is not None) else self._seed
        self._cur_sample = np.uint32(seed) if (seed is not None) else self._seed
```

```
import numpy as np
from UniformGenerator import UniformGenerator
```

```
LM_CONST = np.float64(1)
```

```
class KsiGenerator:
```

```
    def __init__(self, lm=None, seed=None, a=None, c=None, m=None):
        self._lm = np.float64(lm) if (lm is not None) else LM_CONST
        self._uni_gen = UniformGenerator(seed, a, c, m)
```

```
    def _get_next_sample(self):
        k = self._uni_gen._get_next_sample()

        return (1/self._lm)*(np.log(k) - np.log(1 - k))
```

```
    def generate_samples(self, n):
        samples = np.empty(n, dtype=np.float64)
```

```
        for i in range(n):
            samples[i] = self._get_next_sample()
```

```
        return samples
```

```
    def reset(self, seed=None):
        self._uni_gen.reset(seed)
```



```

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import logistic

from KsiGenerator import KsiGenerator

## Model parameters

lambd = 23.45

a = None
c = None
m = None
seed = None

numb_random_variable = 1000
numb_points_for_analitic = 1000

## scipy realisation
ksi_samples_scipy = logistic.rvs(size=numb_random_variable, loc=0, scale=(1/lambd))

## My Generation

ksi_gen = KsiGenerator(lambd, seed, a, c, m)
ksi_samples_my = ksi_gen.generate_samples(numb_random_variable)

## Analytic function

def get_ksi_prob_density(x):
    return lambd*np.exp(-lambd*x)/(np.power((1+np.exp(-lambd*x)), 2))

x = np.linspace(np.min(ksi_samples_scipy), np.max(ksi_samples_scipy), numb_points_for_analitic)
ksi_prob_density_analytic = get_ksi_prob_density(x)

## Plotting

plt.figure()

plt.plot(x, ksi_prob_density_analytic, 'o', label='Analytic func Ksi')
plt.hist(ksi_samples_scipy, density=True, bins='auto', alpha=0.6, label='scipy Ksi')
plt.hist(ksi_samples_my, density=True, bins='auto', alpha=0.6, label='my Ksi')

plt.xlabel('Random variable')
plt.ylabel('Probability')
plt.title("Probability density, lambda = " + str(lambd))
plt.legend()
plt.grid(True)
plt.show()

```

Probability density, $\lambda = 23.45$

